AISD | UFCEP6-0-3 · SECOND RELEASE

Hospital Patient Pathway

SECOND RELEASE · 19 OCTOBER 2026

One executable Camunda 8 process across referral, care, funding and change — revised after peer feedback, with the gaps that remain stated openly.

Hao Zheng (Ruby) 23084382 · Geping Chen (Estrella) 23084393
Minghao Qin (Ender) 23084402 · Guanyan He (Ryan) 23084395

Initial release submission-2026-09-29 (f6fec99)
Current tree f739982 · tagged submission-2026-10-20 · group 10 · draft for second-owner review

Two releases, one process: a short, factual delta

The second release changed the model’s vocabulary and the forms. It did not change assignment, timers, or the coverage profile.

AREA	29 SEPTEMBER · INITIAL RELEASE	19 OCTOBER · SECOND RELEASE
Operational model	Hospital_All_Processes_Simple_C8. One model, one release tag.	New file, Hospital_All_Processes_V3_C8. The substantive change from the first release is the visit split and the slot-skip gateway; twelve pass-through Join_* merges were added for readability. The initial-release file is untouched.
Visit outcome	One outcome value carried booking and attendance. Cancellation, decline and DNA had no separate path.	Six-way VisitOutcome, with AwaitAttendance, RecordDecline and RecordDna; VisitNeedsSlot skips the slot check.
Forms	11 forms. Dates and amounts were entered as free text or loose controls.	12 forms: await_attendance is new; all 12 were rewritten with input validation on 19 October. book_visit gains attendedOn.
Finance rules	Refund decision was coarse: resolved or pending. No full-versus-partial amount control.	Full, partial, transfer, retain or pending; adjustmentAmount is mandatory; an invalid amount fails the job once.

WHAT DID NOT CHANGE

The strategic model is byte-identical to 29 September. All 22 user tasks are still assignee="demo", with zero candidate groups. The model has no timer event of any kind. The cases were re-run on 19 October and the profile did not move.

THE VISIT-OUTCOME SPLIT

Six outcomes where the first release had one path

Cancellation, decline and did-not-attend now have their own outcomes — the reason the pathway decision can be made correctly.

Attended	Booked	Pending / retry	Patient cancelled	Declined	Did not attend
patient came	awaiting the visit	gateway default	no slot needed	no slot needed	no slot needed

WHAT THE MODEL NOW DOES

VisitOutcome routes on six conditions; pending / retry is the default.

AwaitAttendance holds a booked visit and returns to the same decision when attendance is recorded.

RecordDecline and RecordDna record the outcome before the pathway decision.

VisitNeedsSlot sends decline, cancellation and DNA past CheckVisitSlot.

WHY THE OLD MODEL COULD NOT ANSWER THE QUESTION

One outcome value carried both booking and attendance, so a cancellation looked like a visit that still needed a slot.

That forced a slot reservation on paths where no slot was wanted, and left the pathway decision with no distinct event to route on.

The split is modelled, but no acceptance case yet runs DNA → pathway decision → Finance, and reschedule is still a re-booking rather than an outcome of its own.

Delivered in V2 (e4b2159, 305fccb) and carried into V3 unchanged in 357528e. Source: written evaluation, sections 2.2 and 3.3; feedback record F-02.

Twelve forms, and what the validation enforces

Constraints sit on the form each task opens. They are format checks, not identity or role checks.

12

deployed forms

11 at the 29 September baseline.

await_attendance was added with the visit split.

All 12 were rewritten in the 19 October validation pass.

Deployed set: Week6 Coursework/forms/. The pre-validation copy remains at Week3 Coursework/hospital-pathway/forms/; every file differs.

DATE	appointmentDate rejects a past date (disallowPassedDates). attendedOn is a separate field and deliberately allows a past date.
WAITING	waitReason is mandatory when the outcome is pending: no_slot, contact_incomplete or external_waiting.
CHARGE	charge_amount is a two-decimal number from 0.01 to 99999.99, required on the patient-payment path.
INSURANCE	insurerReference is required on the insurer path; its pattern allows letters, digits, hyphen, underscore or slash.
REFUND	adjustmentAmount is required unless the decision is pending: 0.01 to 99999.99, and a partial refund must be below the original charge.

> A format check is not an access check. patientId, staffRole and clinicianId require a leading letter, which rejects a numeric-only value. staffRole is still free text, so the form cannot prove who the user is — the reason BR-37 stays Unsupported.

TRACEABILITY: DID THE GAPS CLOSE?

The gaps narrowed; the coverage barely moved

42 business rules graded against V3 on 19 October: 13 Supported, 25 Partial, 4 Unsupported. One grade changed.

13 SUPPORTED
was 12 · +1

25 PARTIAL
was 26 · −1

4 UNSUPPORTED
unchanged

THE ONE RULE THAT MOVED · BR-20

Refund decision is now full, partial, transfer, retain or pending, with a mandatory amount.

The form and the worker both changed, so the gap the matrix named is closed.

Limits kept in view: no acceptance case reaches FinanceAdjustment — none of the seven on 19 October did — and “authorised Finance member” is still graded under BR-37.

THREE NARROWED, SAME GRADE

BR-36 visit split delivered; no DNA → Finance case, and reschedule is still not its own outcome.

BR-35 booking structured; no out-of-timeframe flag or pathway notification.

BR-13 amount and reference structured; limitations are still free text in notes.

The release improved the process. It did not improve its alignment profile: 26 of the 42 rules are untouched.

WHY 26 RULES DID NOT MOVE

Two missing capabilities carry nine of 26 rules

Grouped by the cause that has to be removed first, not by symptom.

CAUSE	RULES	RULE IDS	WHAT THE CAUSE IS
C1 · No role model	5	BR-09, 19, 21, 27, 31	All 22 user tasks are assignee="demo"; no candidate groups, claims or login.
C2 · No timers	4	BR-06, 32, 33, 34	Zero timer events, so no rule about “within seven days” can be enforced.
C3 · Records unstructured	8	BR-01, 04, 05, 08, 23, 25, 28, 30	Records are still free text in notes, or have no field at all.
C4 · Never exercised	1	BR-22	Change path, unauthorised skip and Finance routing exist; no case drives them.
C5 · Rule undecided	1	BR-29	Urgency highlight rules are still a stakeholder decision; only manual priority exists.
C6 · Never built	7	BR-10, 37, 38, 39, 40, 41, 42	Each is a new capability, not a refinement of an existing one.

C1 and C2 are one capability each with several dependants: one role model unlocks five rules; one timer pattern unlocks four. C3 is the largest group and the cheapest — eight rules need a form field and a check, not a redesign. Only BR-22 is blocked on test coverage alone.

Seven received points, six rows, one left open

Peer feedback only: four items from Group 10, three from the whole-class sheet. No tutor feedback has been received.

ID	POINT RAISED	DECISION AND ACTION	EVIDENCE
F-01	Alternative paths met informally; add explicit merges	Kept the split paths; added 12 pass-through Join_* gateways, seven in V2 and five more in V3.	V3 model
F-02	Decline, cancellation and DNA treated as slot bookings	Six-way VisitOutcome; AwaitAttendance, RecordDecline and RecordDna; slot check skipped on those paths.	V3 model
F-08	Dates and amounts should use constrained controls	Past dates rejected; mandatory waitReason; bounded charge_amount; insurerReference on the insurer path.	Forms, 19 Oct
F-09	Dates must not allow an unreasonable or past date	disallowPassedDates on appointmentDate; requested period bounded to 1–365 days or 1–52 weeks.	book_visit, register_request
F-10	Store data in a database, not only process variables	H2 file database with JPA entities and repositories; workers write audit, booking and payment tables.	application.yaml, domain/
F-11	Enforce role-based access in Camunda	Not implemented. No candidate groups or role login; AT-03 is still a recorded failure.	Record sec. 3

Seven received items map to six rows because Group 10’s items 1 and 2 share F-08. The record’s other five rows (F-03 to F-07) are the group’s own review points, not outside feedback. Tutor rows are blank because no tutor point exists in writing.

ACCEPTANCE RUN · 19 OCTOBER 2026

Seven cases on V3: six pass, one fails

The same seven cases that ran on 29 September ran again against V3_C8. Six pass; AT-03 still fails.

CASE	WHAT IT DRIVES	29 SEPTEMBER	19 OCTOBER · V3	INSTANCE KEY
AT-01	Main path: accept, book, reach clinical care	PASS	**PASS**	2251799813930310
AT-02	Referral rejected by the consultant	PASS	**PASS**	2251799813930247
AT-03	Medical Secretary tries to complete Accept	FAIL	**FAIL**	2251799813930421
AT-04	No suitable slot: booking reopens as pending	PASS	**PASS**	2251799813930493
AT-05	Payment fails, then is retried	PASS	**PASS**	2251799813930607
AT-06	Payment may be taken; confirmation missing	PASS	**PASS**	2251799813930876
G07	Full route to discharge, nine human tasks	PASS	**COMPLETED**	2251799813931091

WHAT SIX PASSES DO NOT PROVE

They prove the model executes the path each case drives, not the branches no case enters. AT-03 fails exactly as it did on 29 September: there are still no candidate groups, so no role is enforced. Underneath it, 19 unit tests pass and 22 user tasks bind to 12 forms.

All results come from the 21:40 session after restarting c8run with --disable-connectors; the 21:38 attempt died on a connection reset and is kept as evidence.

EVIDENCE AND LIMITS

What the run proves, and what it does not

Seven executions, one failure, four boundaries no case crosses.

PASS

G07 COMPLETED · six of seven

Nine human tasks, zero active incidents: instance 2251799813931091 reached COMPLETED on V3.
AT-01, AT-02, AT-04, AT-05 and AT-06 pass, as they did on 29 September.
19 unit tests pass, and 22 user tasks bind to 12 form ids with no orphan.

NOT COVERED

What a REST driver cannot test

Values are posted through /v2, so browser-side validation is untested: required fields, patterns, disallowPassedDates.
The role layer is untested because it does not exist.
No timer exists in the model, so no timer behaviour is tested.

FAIL

AT-03 · the role boundary

A Medical Secretary completing Accept still created a BookVisit.
The same defect in both releases, not a regression: no candidate groups exist, so no role can be enforced.
It becomes a pass only when BR-37 is delivered.

OPEN

Branches no case drives

The formal change path, DNA to payment, urgent authorisation, the refund adjustment, letter escalation.
Modelled, not exercised: no V3 result is claimed for any of them.

A pass here means one path executed, not that the rule behind it is satisfied: the coverage profile stays 13 / 25 / 4.

PLAN VERSUS ACTUAL

What was planned, and what actually shipped

Read against the 16 October plan; status checked on 19 October.

SPRINT 3

Planned

PB-30 improved operational BPMN and readable export; PB-16 second-owner sign-off; PB-34 V1-to-V2 change note.

Actual

Model delivered as V2, then renamed and extended as V3.

Readable operational export produced on 19 October, after the plan was written.

PB-34 done. PB-16 still unsigned: both second-owner reviews are open.

SPRINT 4

Planned

PB-31 verify the revised solution and name the commit behind every result; PB-39 carry the traceability grades forward.

Actual

PB-39 done: 13 / 25 / 4 recorded against V3, with the rule-by-rule reason.

PB-31 verified: seven cases re-run on V3 on 19 October: six pass, AT-03 fails.

The plan’s environment risk showed once: the first c8run start died on the connectors health check; the run was repeated with connectors disabled.

WHAT THE RECORD SHOWS

Process records were left to the end: Standup 3 has no record, Standup 4 has one row of four, SR3’s fields are blank and SR4 has no date. The deck itself was still listed as “not started” in the plan.

Consequence: the model and the forms moved forward while the evidence and the records stayed at 29 September. That is the gap this deck has to make visible.

LEVERAGE ORDER FOR THE NEXT RELEASE

Five items, ordered by rules moved per unit of work

Measured by rules moved per unit of work — written evaluation, section 3.7.

1	**BR-37 — role model and candidate groups** Unlocks BR-09, BR-19, BR-21, BR-27 and BR-31, and turns AT-03 from a recorded failure into a pass condition. No other item comes close.	P0
2	**BR-38 — audit immutability** The append-only writer exists; a read-only audit API and a refusal test do not. Also underwrites the enquiry trail in BR-28.	P0
3	**BR-08 — consent and a structured treatment request** The artefact did not change at all in this release: the task text asks for fields the form does not have.	P0
4	**The timer pattern — one implementation, four rules** BR-06, BR-32, BR-33 and BR-34 all wait on the same missing capability. One pattern applied at four places.	C2
5	**BR-22 — three acceptance cases** The formal change path, the unauthorised skip and Finance routing already behave; only the cases are missing.	Cheapest

LIVE DEMONSTRATION

A repeatable route through V3_C8

Start Camunda first, then the workers from Week3 Coursework/hospital-pathway.

01 REFERRAL AND BOOKING

Register → accept → book → attend

register_request → review_referral (accept) → book_visit (booked) → await_attendance (attended).
Watch visitOutcome change, then resolve to attended.

02 CARE AND FUNDING

Care → letter → self-pay 10.01

clinical_care → dispatch_letter (sent) → funding_assessment (patient payment, 10.01) → request-payment publishes payment-result.
Watch payment_status = successful.

03 FINISH

Resources → review → discharge

book_treatment returns resources ready → send-booking-confirmation → back to clinical_care for the post-treatment review.
The care-plan gateway then routes Discharge to EndCare.
Watch booking_slot_status = CONFIRMED and the instance reach COMPLETED.

WHAT TO WATCH, AND WHAT NOT TO CLAIM

Start Hospital_All_Processes_V3_C8; the build packages the 12 forms from Week6 Coursework/forms/. A decline or DNA takes VisitNeedsSlot past CheckVisitSlot — that is the changed behaviour worth showing. This is a demonstration; the acceptance evidence is the 19 October run, not a walk-through.

REFERENCES AND LINKS

Every claim resolves to an openable path

Repository paths are relative to the root at f739982; each is linked to the GitHub copy.

WRITTEN EVALUATION

submissions/BPMEA/BPMEA_Written_Evaluation.md

TRACEABILITY MATRIX

docs/requirements-traceability-matrix.md

FEEDBACK RESPONSE RECORD

Week6 Coursework/feedback/feedback-response_2026-10-16.md

OPERATIONAL MODEL V3

Week6 Coursework/bpmn/Hospital_All_Processes_V3.bpmn

DEPLOYED FORM SET

Week6 Coursework/forms/ (12 files)

V3 RUN AND BUILD, 19 OCT

Week3 Coursework/hospital-pathway/evidence/Camunda_V3_Acceptance_2026-10-19.txt

INITIAL-RELEASE ACCEPTANCE

submissions/AISD/Group_Acceptance_Test_Plan_and_Results_2026-09-29.pdf

INITIAL-RELEASE DECK

submissions/BPMEA/Hospital_Pathway_BPMEA_Presentation_2026-09-29.pptx

SPRINT 3–4 PLAN

Week6 Coursework/plan/sprint3-4-plan_2026-10-16.md

STANDUP AND REVIEW RECORDS

Week6 Coursework/process/

REPOSITORY

github.com/Estrella6066773/HW_AISD

RELEASE COMPARISON

github.com/Estrella6066773/HW_AISD/compare/f6fec99...f739982

Initial release submission-2026-09-29 = f6fec99 · Second release tree f739982 · Group 10, AISD UFCEP6-0-3